

Facilitating Joint Awareness Within Care Networks for Noise Sensitivity Management and Regulation

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ABSTRACT

Noise sensitivity impacts many autistic and non-autistic people, making self-regulation challenging and impacting their quality of life. My doctoral research aims (1) to understand the experiences of people with noise sensitivity (PWNS); (2) to identify how PWNS and their allies become aware of, manage, and regulate responses to noise sensitivity; and finally, (3) to design and evaluate the feasibility of technology to facilitate awareness amongst PWNS and their care networks. Completed studies thus far have informed the design of a multi-device application to support noise sensitivity. Focusing on the concept of joint awareness, I am currently working on a deployment study to evaluate the feasibility of our app to facilitate awareness within care networks for noise sensitivity management and regulation. My dissertation research will provide an understanding of noise sensitivity experiences, introduce the concept of joint awareness, and provide technological solutions to support collaborative management and regulation of noise sensitivity.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing devices**.

KEYWORDS

Noise Sensitivity, Assistive Technology, Collaborative Technology

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1 INTRODUCTION

Noise sensitivity is a phenomenon associated with sensory processing disorder and over half of autistic people and a large portion of non-autistic people report experiencing it [2, 24]. Self-regulation under these conditions can be challenging, leading to responses ranging from fleeing an area to self-injury [14, 22]. Strategies like sound masking and avoidance behaviors are suboptimal, impacting situational awareness, comfort, stigma, cost, and dependency

[16], underscoring the need for better regulation and management practices. Within the field of Human-Computer Interaction (HCI), research on sound, sensitivities, and/or awareness has so far been mostly isolated to the D/deaf and hard-of-hearing community to support sound awareness and recognition [6, 11–13]. In the autism space, research has largely focused on supporting children through the design of inclusive environments, stress or anxiety reduction, and assessment methods for sensory differences [5, 15, 18]. Indeed, others have investigated regulation practices through wearables, virtual worlds, and online gaming [1, 3, 20, 21, 23]. However, these approaches of support focus on the individual and how they self-regulate. Yet, prior literature highlights the importance of collaborative care for successfully managing health and well-being long-term [4, 19]; therefore, it is important for people with noise sensitivity to not only be aware and manage their health alone but to also have others who are aware to support in managing when needed. Thus, my work builds on prior work by focusing on facilitating awareness between people with noise sensitivity (PWNS) and their care networks and supporting management and regulation strategies through technology.

2 RESEARCH APPROACH

My doctoral research expands on this prior work by examining experiences of noise sensitivity and technology-mediated approaches for supporting awareness, management, and regulation practices between PWNS and their care networks. Thus, I have framed my research around the following questions:

- RQ1: How do PWNS discuss noise sensitivity and the related challenges they experience?
- RQ2: How do PWNS and their allies become aware of, manage, and regulate responses to noise sensitivity?
- RQ3: How can technology be designed to facilitate awareness and support amongst PWNS and their care networks?

To address these questions, over the past three years, I have studied how people make sense of their noise sensitivity, the role of awareness and collaboration in management, and the challenges they consider when designing technology to support their management and regulation needs through three studies (see Figure 1).

2.1 Understanding the Experiences of PWNS

In collaboration with other researchers, we analyzed interactions among autistic people in online forums as they discussed and supported each other regarding their described noise sensitivities [10]. This analysis revealed three themes: sharing experiences of noise sensitivity, managing noise sensitivity, and disclosing autism. Key findings show several societal barriers and challenges impacting

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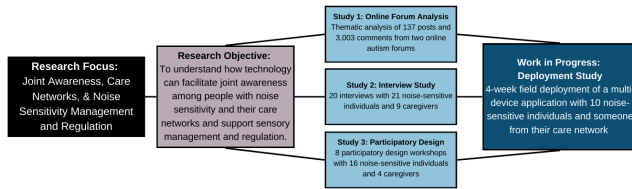


Figure 1: Dissertation Overview

noise sensitivity management and coping preferences, ranging from inaccessible environments to a lack of home, school, and workplace accommodations. Taking note of current management practices, we provided design implications for wearable and sensor technology, identifying that such technology can be used to monitor environmental and physiological climates to send feedback to users to increase awareness and provide personalized approaches for managing noise sensitivity based on certain contexts. This work informed the preliminary design of our assistive application, AudioBuddy [8]. The initial version of AudioBuddy was developed for WearOS and Android devices. It included notifications of increased noise levels, supported self-regulation and emotion regulation principles from the Zones of Regulation¹ [17], and educational materials for self-advocacy.

2.2 Accessibility Through Awareness

The second study conducted was an interview study to explore the management and regulation practices of PWNS and their care networks. Key findings suggest that to support regulation during noise sensitivity experiences, the person with noise sensitivity or someone with enough influence to respond and manage must be aware that 1) the sensitivity exists and 2) something such as a sound, person, or environment is triggering that sensitivity. This awareness can be more general, in the moment, after the event, or some combination. Finally, this work introduces the concept of joint awareness, which fosters collaborative management and regulation of noise sensitivity.

2.3 Co-designing with PWNS and Their Care Networks

My latest work used participatory design to inform the design of technologies to support PWNS and their care networks. Key findings emphasized the value of sensing and tracking technologies, such as applications to monitor body temperature, step count, blood pressure, and location, to increase awareness of individuals' well-being. Participants expressed a desire not only for data tracking but also for presenting this data to facilitate reflection and comprehension for themselves and with those in their care network. Customization of tools to offer context-specific self-regulation strategies was highlighted as essential, with the design accommodating family involvement in coping strategies, which was particularly noted among parent designers. Finally, along with collaborators, I identified inclusive practices for involving children with noise

¹The Zones of Regulation is a social emotional learning curriculum designed for self-regulation and emotion regulation that teaches how to manage and regulate their feelings. More information can be found at <https://zonesofregulation.com/>

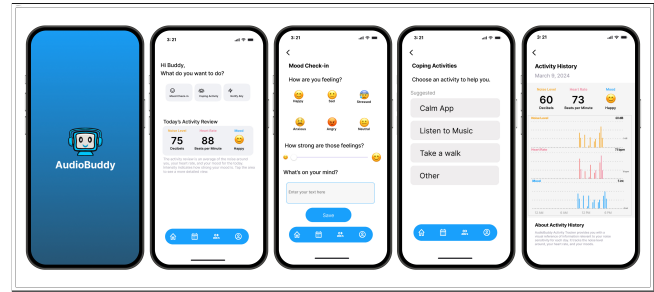


Figure 2: AudioBuddy System. 1) Splash screen, home screen, mood check-in and reflection, coping activities, and activity history showing the day's noise level, heart rate, and mood data.

sensitivity in the design process [9], which had further implications in child-centered AI design and testing [7]. In summary, this latest study indicates a strong avenue for wearable and ubiquitous technologies, specifically regarding opportunities for sensing and tracking, personalization, and self- and co-regulation.

2.4 Designing AudioBuddy

The aforementioned studies have led to the latest redesign of AudioBuddy (Figure 2). It operates in two modes: an event-triggered scenario where the application sends a notification based on tracked physiological and environmental data, prompting the users to engage in the app, or a manual mode through independent engagement. AudioBuddy includes a toolbox of regulation strategies like breathing exercises, music listening, and physical activities, with users rating the effectiveness of each tool for a more personalized experience. The system also facilitates awareness through emotion check-ins, journaling, and data reflection. Finally, users can share their information with someone in their care network to increase their awareness and offer opportunities to engage in collaborative management and regulation strategies when needed.

3 WORK IN PROGRESS: DEPLOYMENT STUDY

I am currently planning the final study for my doctoral research, where I, along with collaborators, will evaluate the feasibility of our app to facilitate awareness within care networks for noise sensitivity management and regulation. To do so, a field deployment with the application over a four-week period will be conducted with 10 PWNS and someone from their care network (20 participants total). At the start of the deployment study, participants will undergo an onboarding interview where they will be provided with an Apple Watch and iPhone with the AudioBuddy system on it. This interview will include a walk-through of the app and an opportunity to ask questions about the process. During the four-week period, participants may receive notifications about environmental or physiological indicators of potential noise sensitivity incidents, prompting them to engage in a regulation activity and complete an in-app survey to document these experiences. At the end of each day, participants will also complete a web-based survey to reflect on their overall experiences. Additionally, thirty-minute weekly interviews will be conducted to review participants' experiences

for the week. Finally, study participants will engage in an interview that summarizes their experiences using AudioBuddy.

4 EXPECTED CONTRIBUTIONS

Multiple novel contributions are anticipated through my dissertation research toward sensory sensitivity research and the HCI/CSCW fields. First, I aim to illuminate the lived experiences of noise-sensitive people and their respective care networks. This insight will further inform inclusive design paradigms for sensory-friendly spatial and technological domains. Second, my dissertation research will contribute innovative frameworks for integrating noise sensitivity stakeholders into user-centered design and development methods. Building upon this empirical foundation, this research will contribute to the design and development of assistive technologies to facilitate sensory sensitivity awareness, management, and regulation practices for support within care networks, thus further establishing the applicability of assistive technology to amend challenges associated with sensory processing disorders. By addressing these crucial aspects, my dissertation research seeks to significantly improve the quality of life for noise-sensitive individuals and their support systems.

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